Utsab Khadka

Backend & Data Engineering | Computer Science Student

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SUMMARY

BSc CSIT student with four semesters completed, focused on backend engineering and data pipelines. Builds APIs and data services in Python with FastAPI and PostgreSQL, writing raw SQL, migrations, tests and error handling by hand to understand how the pieces work. Also exploring AI engineering through LLM-based structured data extraction.

EDUCATION

BSc in Computer Science and Information Technology (BSc CSIT) Expected Nov 2028

Tribhuvan University – Central Campus of Technology, Dharan, Nepal | **GPA:** 3.16/4.0 (through semester 3 of 4 completed)

Relevant coursework: Data Structures and Algorithms, Database Management System, Operating Systems, Computer Networks, Artificial Intelligence, Object Oriented Programming, Discrete Structure, Theory of Computation, Computer Architecture, Numerical Methods, Statistics I & II

TECHNICAL SKILLS

Languages: Python, SQL
Backend: FastAPI, Pydantic, REST API design, async Python, httpx, Requests
Databases: PostgreSQL (raw SQL via psycopg, migrations, constraints, indexing)
AI / LLM: Structured LLM extraction (instructor, litellm), prompt-to-schema pipelines
Tools: Git, GitHub, Linux (basics), pytest, Locust, Render, NeonDB
Learning: Docker, Redis, microservices, PostgreSQL query optimization

PROJECTS

DataPump – Data Ingestion Pipeline GitHub | in progress

Python, FastAPI, PostgreSQL, psycopg (async), Pydantic

- Building a data-ingestion service on FastAPI and PostgreSQL, using an async connection pool and hand-written SQL (no ORM); 95+ commits so far.
- Designed a versioned migration with a status enum, check constraints that keep job status and progress fields consistent, and a composite index on (status, created_at).
- Added a seed script that generates constraint-valid test data, plus a custom database exception hierarchy mapped to HTTP status codes (408, 409, 422, 503) through a global handler.

GraphCrawl AI – AI-Powered Web Data Extraction GitHub

Python, httpx, BeautifulSoup, instructor, litellm, Pydantic, pytest

- Built an async Python library that turns a webpage URL into typed Pydantic data: fetch with retries and timeouts, HTML cleaning, then LLM extraction through instructor and litellm; 172+ commits.
- Provider-agnostic design (Gemini, OpenAI or Anthropic models via litellm); built-in extraction modes for page summaries, contact details and product listings. Verified against Gemini models.
- Wrote a pytest suite with mocked HTTP (respx) covering input validation, fetching and HTML cleaning, and a typed exception hierarchy for crawl, fetch and LLM errors.

Framework-less Library API GitHub | Live API

Python (stdlib http.server), PostgreSQL, psycopg, Render, NeonDB, Locust

- Built a REST CRUD API for authors and books using only Python's standard library, with hand-written routing, SQL migrations and a seed script (500+ books); deployed live on Render with NeonDB.
- Load-tested with Locust and documented where it breaks (unstable at roughly 35–40 concurrent requests), with thread pooling and async I/O planned as fixes.

Weather API Data Fetcher GitHub

Python, Requests, Open-Meteo API, logging, CSV

- CLI tool that pulls live weather data with input validation, request retries, rotating logs and CSV storage.